

Integrated Circuits: Introduction, Fabrication and Limitations

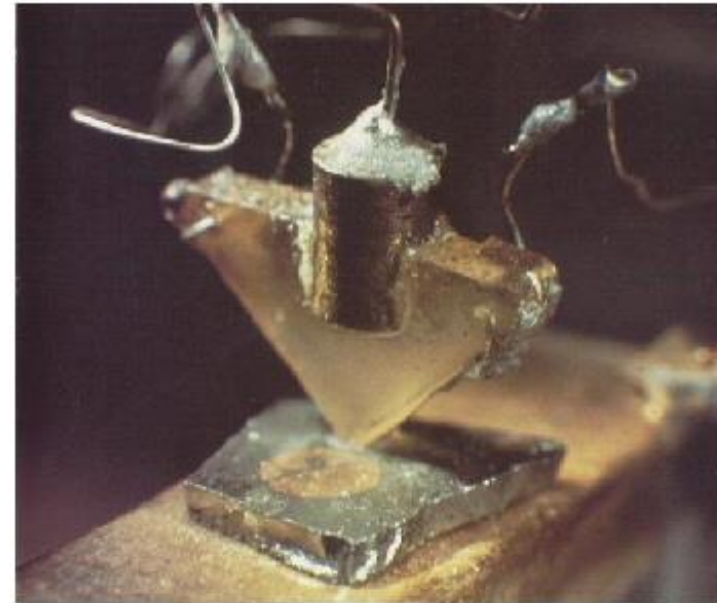
By

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A Brief History

Invention of the Transistor

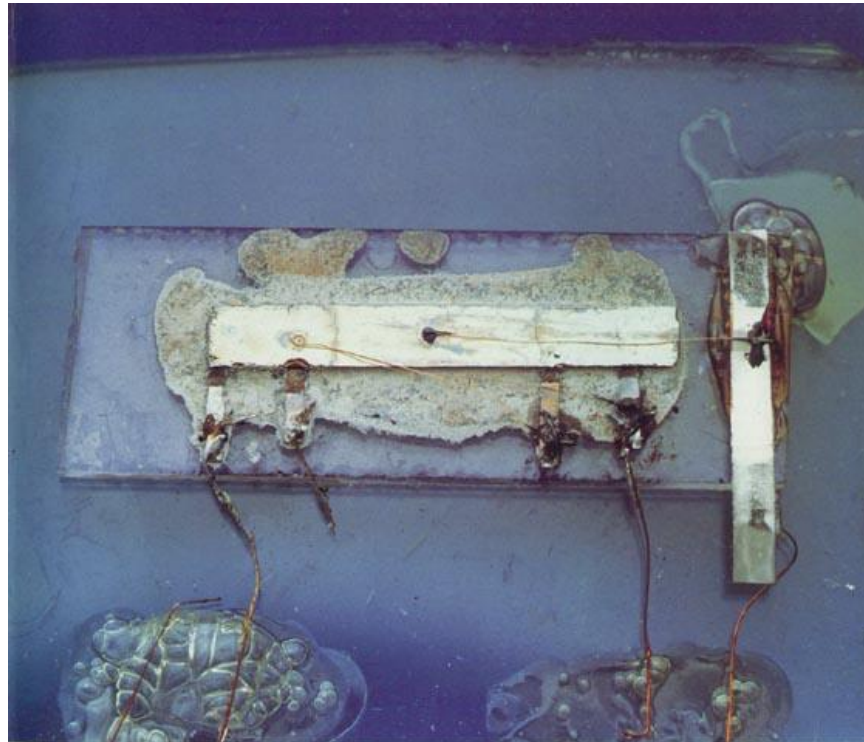
- Vacuum tubes ruled in first half of 20th century Large, expensive, power-hungry, unreliable
- 1947: first point contact transistor (3 terminal devices)
 - Shockley, Bardeen and Brattain at Bell Labs



A Brief History, contd..

❑ 1958: First integrated circuit

- Flip-flop using two transistors
- Built by Jack Kilby (Nobel Laureate) at Texas Instruments
- Robert Noyce (Fairchild) is also considered as a co-inventor

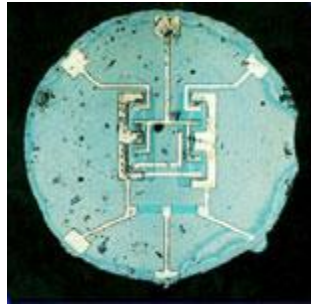


Kilby's IC

smithsonianchips.si.edu/augarten/

A Brief History, contd.

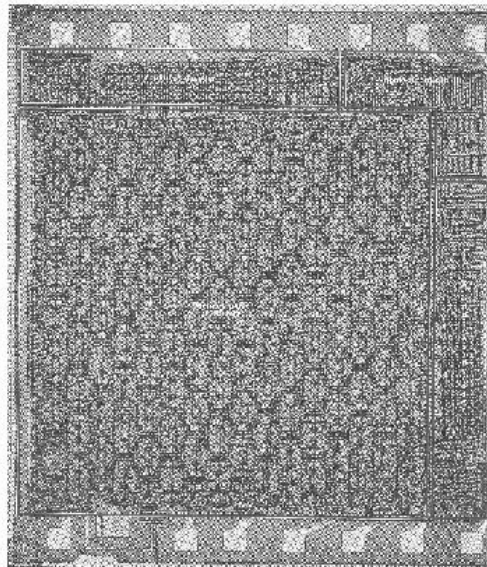
- First Planer IC built in 1961



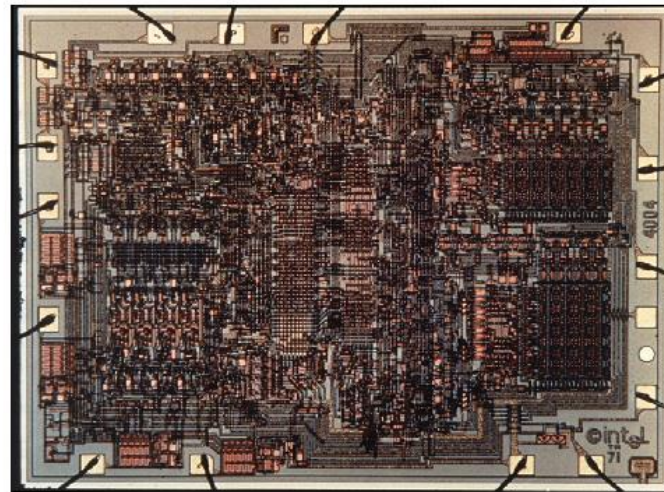
- 2003
 - Intel Pentium 4 μ processor (55 million transistors)
 - 512 Mbit DRAM (> 0.5 billion transistors)
- 53% compound annual growth rate over 45 years
 - No other technology has grown so fast so long
- Driven by miniaturization of transistors
 - Smaller is cheaper, faster, lower in power!
 - Revolutionary effects on society

MOS Integrated Circuits

- ❑ 1970's processes usually had only nMOS transistors
Inexpensive, but consume power while idle
- ❑ 1980s-present: CMOS processes for low idle power



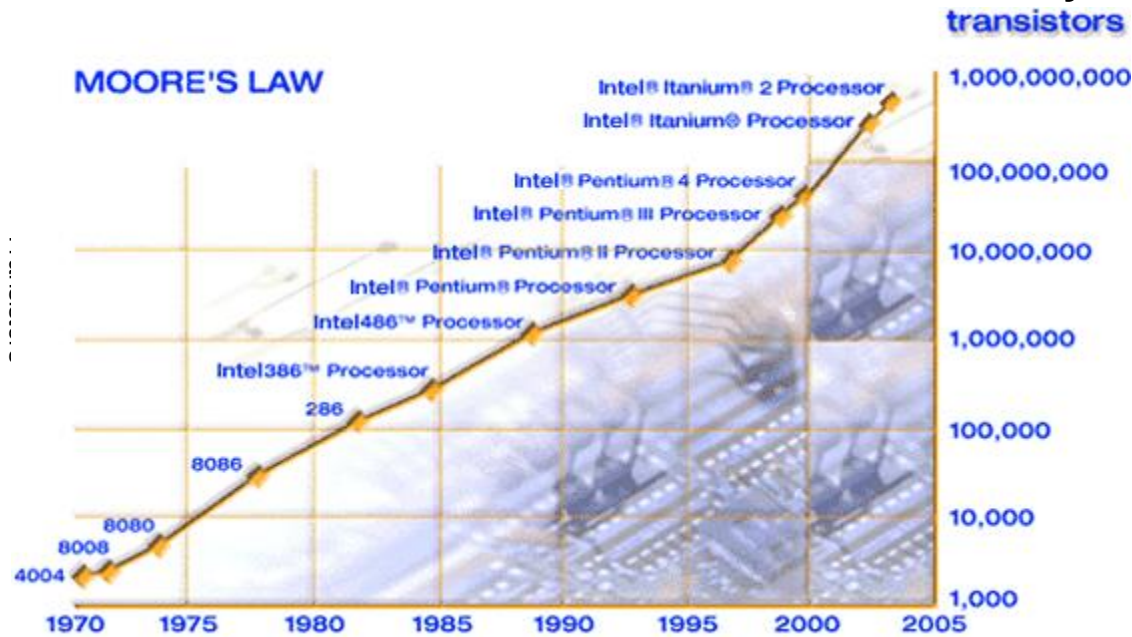
Intel 1101 256-bit SRAM



Intel 4004 4-bit μ Proc

Moore's Law

- 1965: Gordon Moore plotted transistor on each chip
 - Fit straight line on semilog scale
 - Transistor counts have doubled every 24 months



Integration Levels

SSI: 10 gates

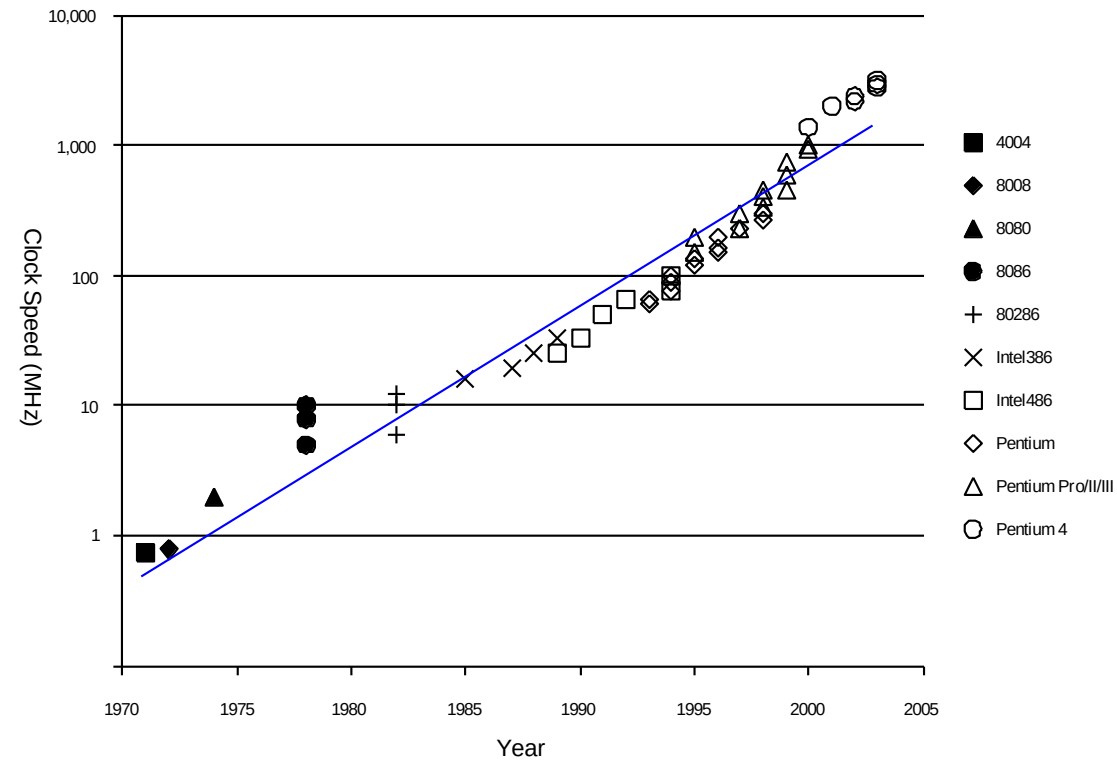
MSI: 1000 gates

LSI: 10,000 gates

VLSI: > 10k gates

Corollaries

- Many other factors grow exponentially
 - Ex: clock frequency, processor performance

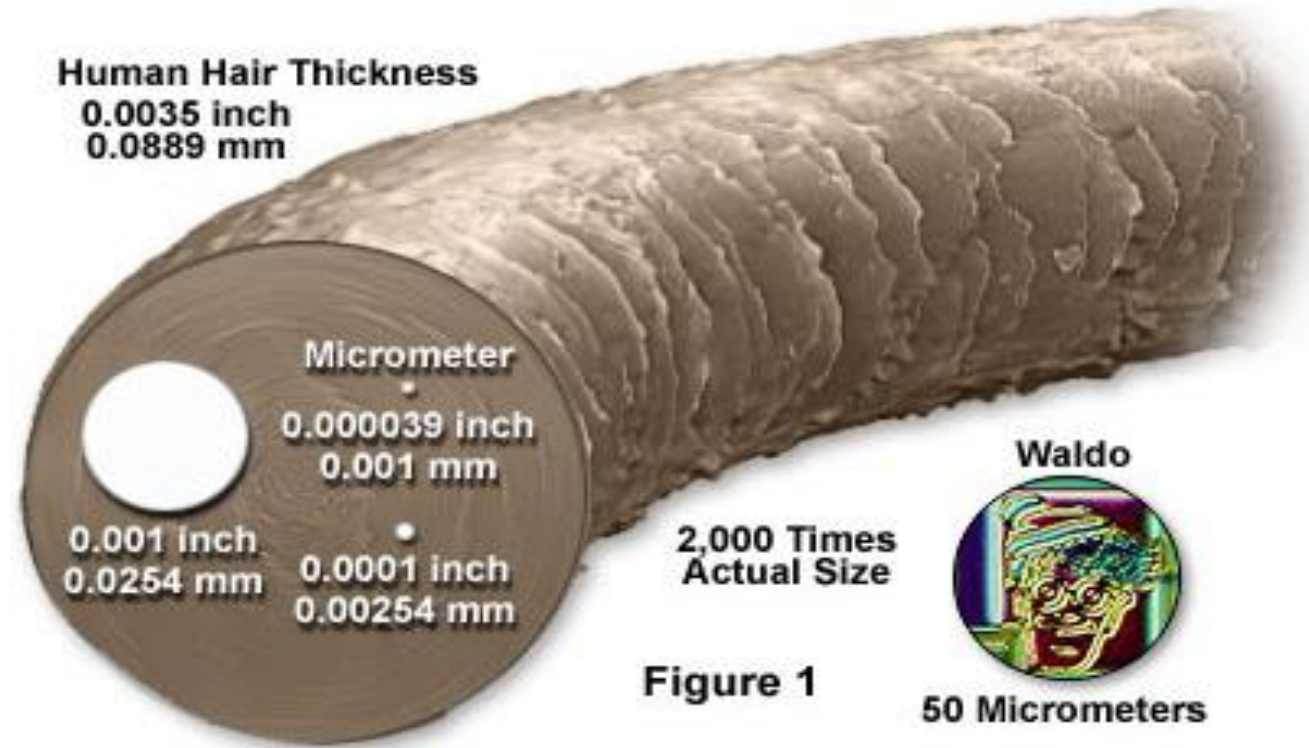


Pentium 4 Processor



http://www.intel.com/intel/intelis/museum/online/hist_micro/hof/index.htm

- Modern transistors are few microns wide and approximately 0.1 micron or less in length
- Human hair is 80-90 microns in diameter



Ref: <http://micro.magnet.fsu.edu/creatures/technical/sizematters.html>

Transistor Types

- Bipolar transistors
 - npn or pnp silicon structure
 - Small current into very thin base layer controls large currents between emitter and collector
 - Base currents limit integration density
- Metal Oxide Semiconductor Field Effect Transistors
 - nMOS and pMOS MOSFETS
 - Voltage applied to insulated gate controls current between source and drain
 - Low power allows very high integration
 - First patent in the '20s in USA and Germany
 - Not widely used until the '60s or '70s

Introduction

- The term CMOS stands for “**Complementary Metal Oxide Semiconductor**”.
- This is one of the most popular technology in the computer chip design industry and it is broadly used today to form integrated circuits in numerous and varied applications.
- This technology makes use of both P channel and N channel semiconductor devices.
- This is the dominant semiconductor technology for microprocessors, microcontroller chips, memories like RAM, ROM, EEPROM and application-specific integrated circuits (ASICs).

- The main advantage of CMOS over NMOS and BIPOLAR technology is the much **smaller power dissipation**.
- Unlike NMOS or BIPOLAR circuits, a Complementary MOS circuit has almost **no static power dissipation**.
- Power is only dissipated in case the circuit actually switches.
- Complementary Metal Oxide Semiconductor transistor consists of P-channel MOS (PMOS) and N-channel MOS (NMOS).

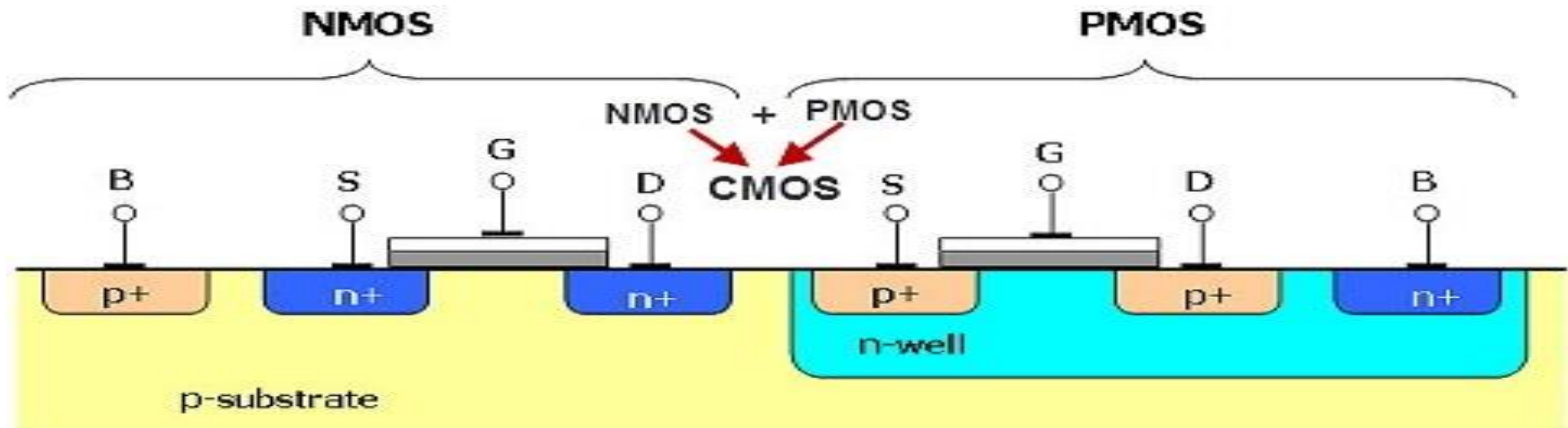


Figure 1. CMOS (Complementary Metal Oxide Semiconductor)

- NMOS is built on a p-type substrate with n-type source and drain diffused on it.
- In NMOS, the majority of carriers are electrons.
- When a high voltage is applied to the gate, the NMOS will conduct.
- Similarly, when a low voltage is applied to the gate, NMOS will not conduct.
- NMOS is considered to be faster than PMOS, since the carriers in NMOS, which are electrons, travel twice as fast as the holes.

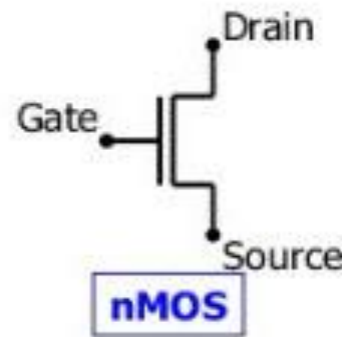


Figure 2. NMOS Transistor

- P- channel MOSFET consists of P-type Source and Drain diffused on an N-type substrate.
- The majority of carriers are holes.
- When a high voltage is applied to the gate, the PMOS will not conduct.
- When a low voltage is applied to the gate, the PMOS will conduct.
- The PMOS devices are more immune to noise than NMOS devices.

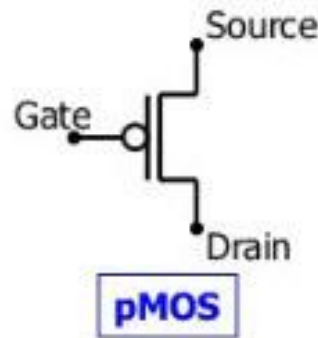


Figure 3. *PMOS Transistor*

- In CMOS technology, both N-type and P-type transistors are used to design logic functions.
- The same signal which turns ON a transistor of one type is used to turn OFF a transistor of the other type.
- In CMOS logic gates a collection of n-type MOSFETs is arranged in a pull-down network between the output and the low voltage power
- CMOS logic gates have a collection of p-type MOSFETs in a pull-up network between the output and the higher-voltage rail (often named V_{dd}).r supply rail (V_{ss} or quite often ground).

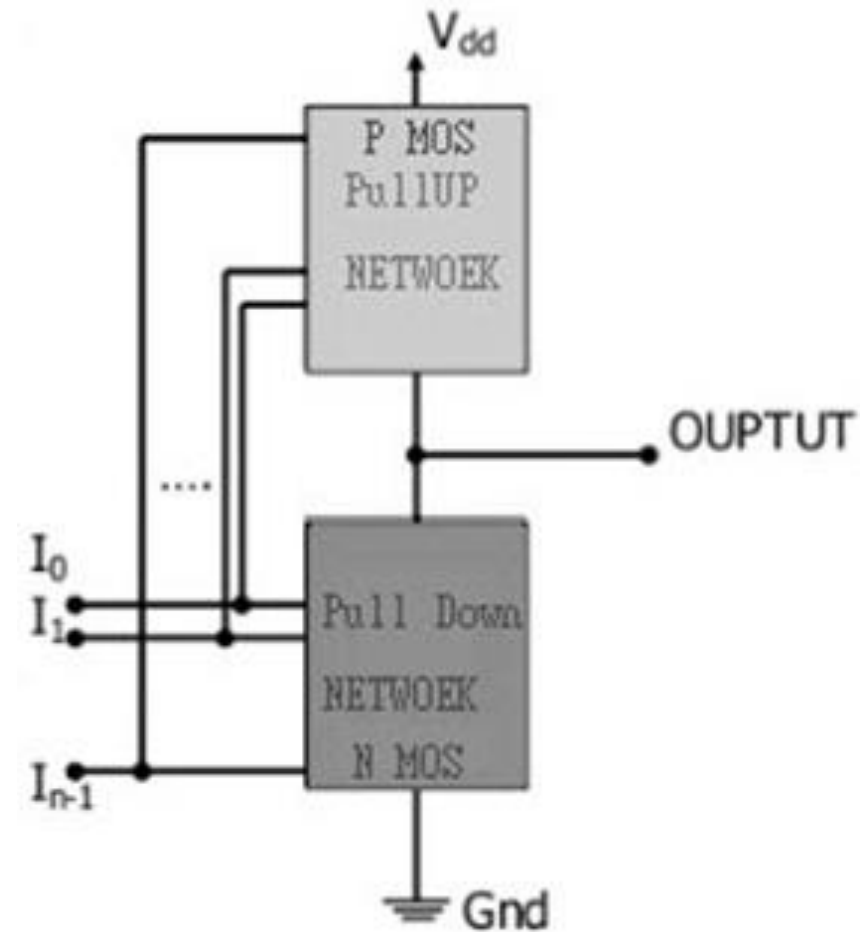


Figure 4. CMOS using Pull Up & Pull Down network

CMOS Inverter

- The input V_{in} serves as the gate voltage for both transistors.
- The NMOS transistor has input from V_{ss} (ground) and the PMOS transistor has input from V_{dd} .
- The V_{out} is output voltage.
- When a high voltage ($\sim V_{dd}$) is given at input terminal (A) of the inverter, the PMOS becomes an open circuit, and NMOS switched OFF so the output will be pulled down to V_{ss} .
- When a low-level voltage ($< V_{dd}$, $\sim 0v$) applied to the inverter, the NMOS switched OFF and PMOS switched ON. So the output becomes V_{dd} or the circuit is pulled up to V_{dd} .

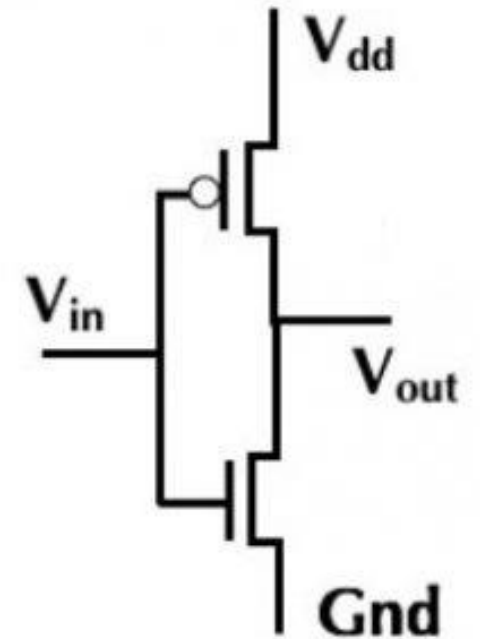


Figure 5. *CMOS Inverter*

VTC of CMOS Inverter

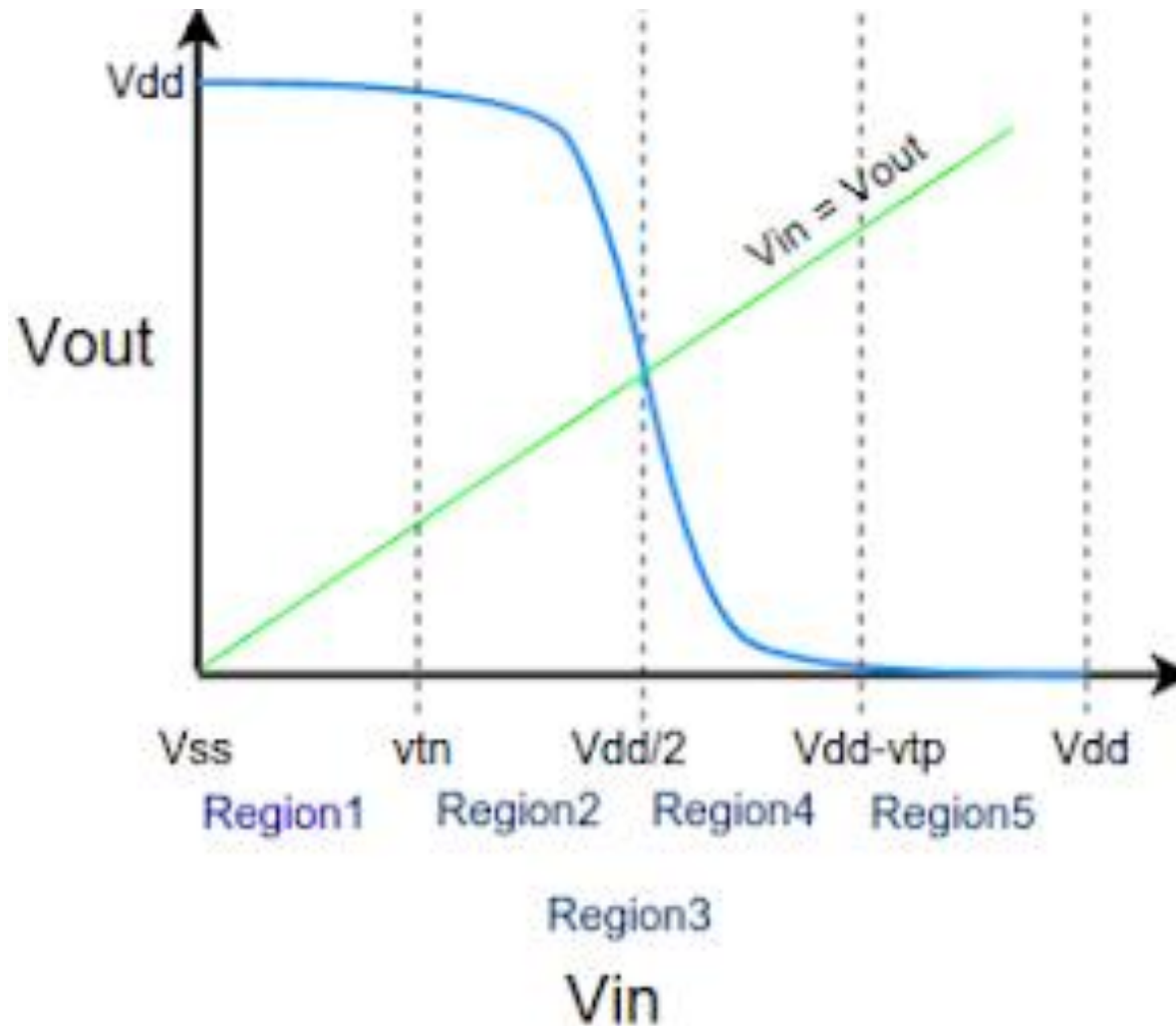


Figure 6. Voltage Transfer Characteristics

Power Supply Voltage

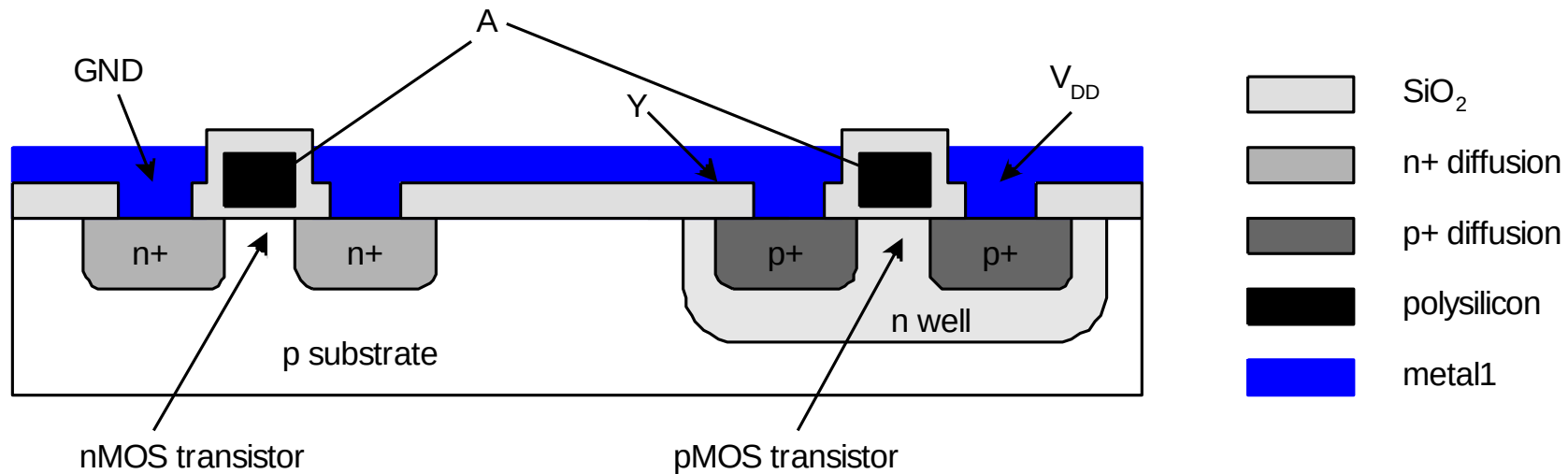
- $GND = 0\text{ V}$
- In 1980's, $V_{DD} = 5\text{V}$
- V_{DD} has decreased in modern processes
 - High V_{DD} would damage modern tiny transistors
 - Lower V_{DD} saves power
- $V_{DD} = 3.3, 2.5, 1.8, 1.5, 1.2, 1.0,$
- Effective power supply voltage can be lower due to IR drop across the power grid.

CMOS Fabrication

- CMOS transistors are fabricated on silicon wafer
- Wafers diameters (200-300 mm)
- Lithography process similar to printing press
- On each step, different materials are deposited, or patterned or etched
- Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process

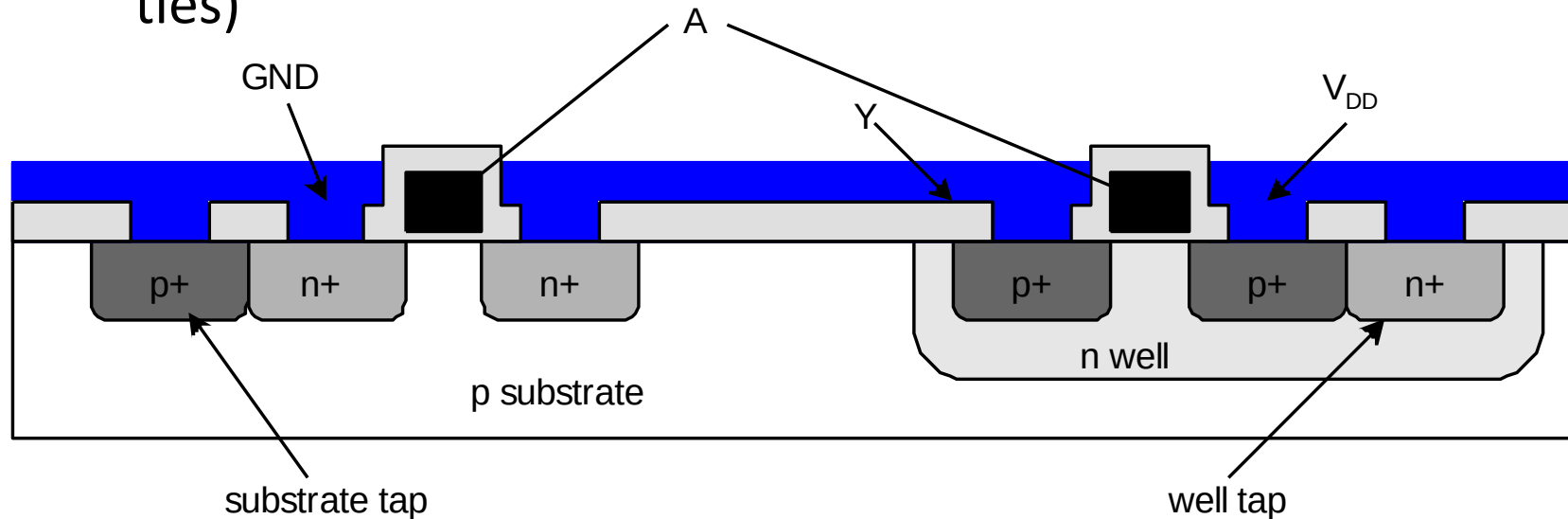
Inverter Cross-section

- Typically use p-type substrate for nMOS transistors
- Requires to make an n-well for body of pMOS transistors



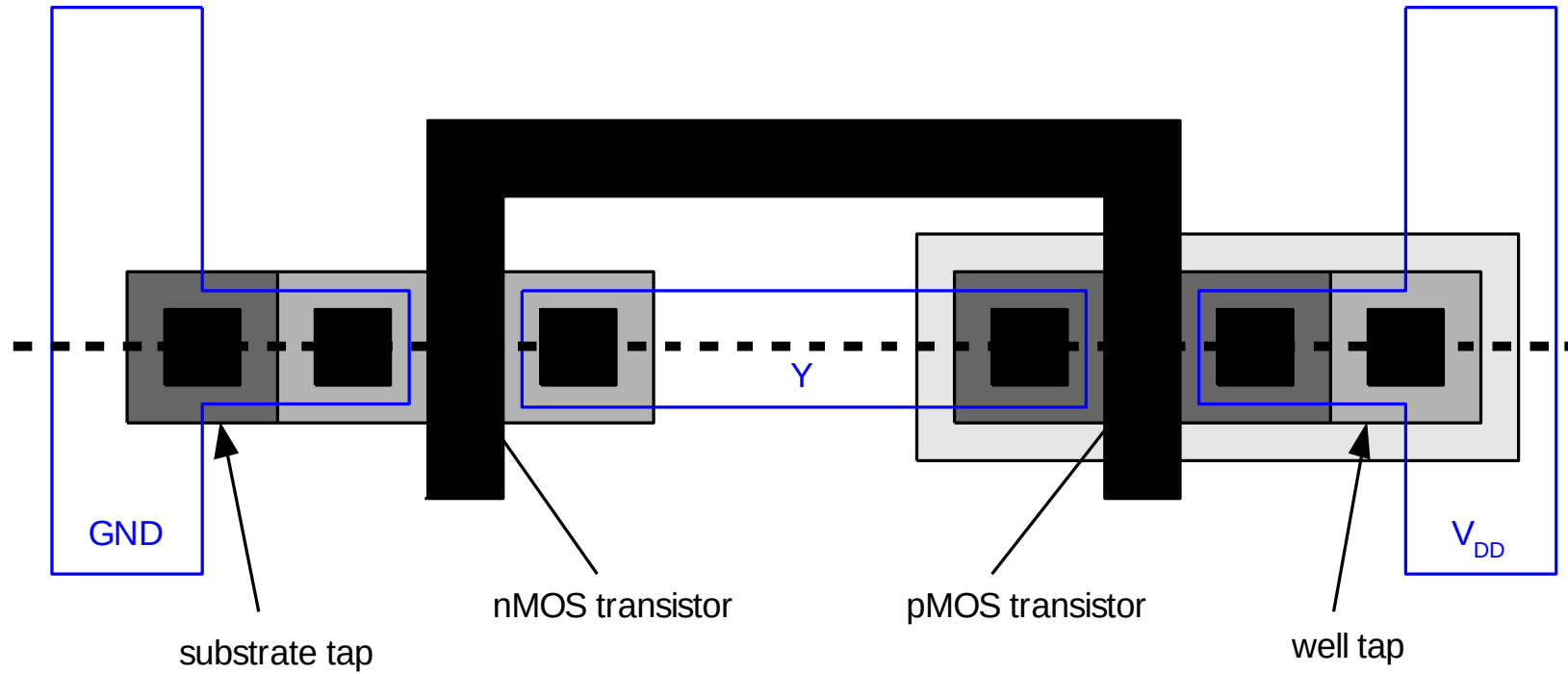
Well and Substrate Taps

- Substrate must be tied to GND and n-well to V_{DD}
- Metal to lightly-doped semiconductor forms poor connection called Schottky Diode
- Use heavily doped well and substrate contacts/taps (or ties)



Inverter Mask Set

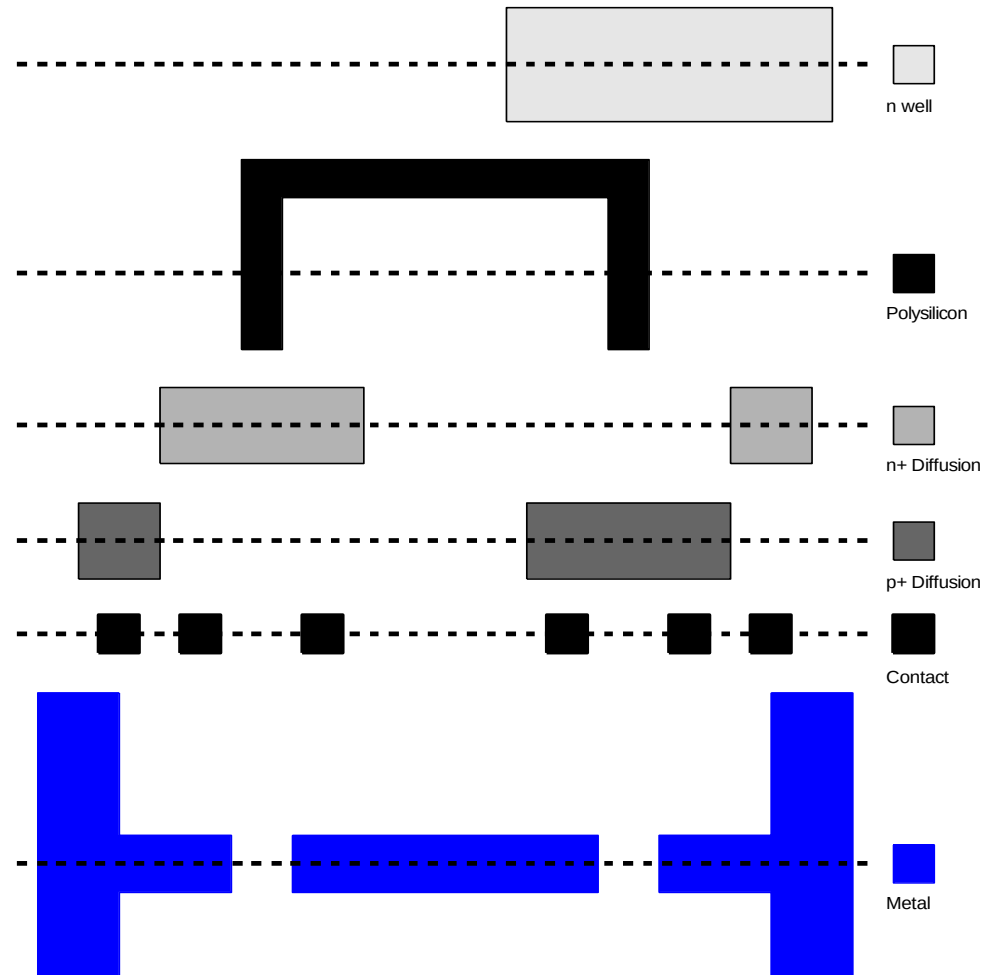
- Top view
- Transistors and wires are defined by *masks*
- Cross-section taken along dashed line



Detailed Mask Views

- Six masks
 - n-well
 - Polysilicon
 - n+ diffusion
 - p+ diffusion
 - Contact
 - Metal

**In reality >40 masks
may be needed**



Fabrication Steps

- Start with blank wafer (typically p-type where NMOS is created)
- Build inverter from the bottom up
- First step will be to form the n-well (where PMOS would reside)
 - Cover wafer with protective layer of SiO_2 (oxide)
 - Remove oxide layer where n-well should be built
 - Implant or diffuse n dopants into exposed wafer to form n-well
 - Strip off SiO_2

1. **Wafer:** The silicon wafer is selected for doping p-type impurities on it. The doped wafer formed will be the p-type substrate with the trivalent impurities.

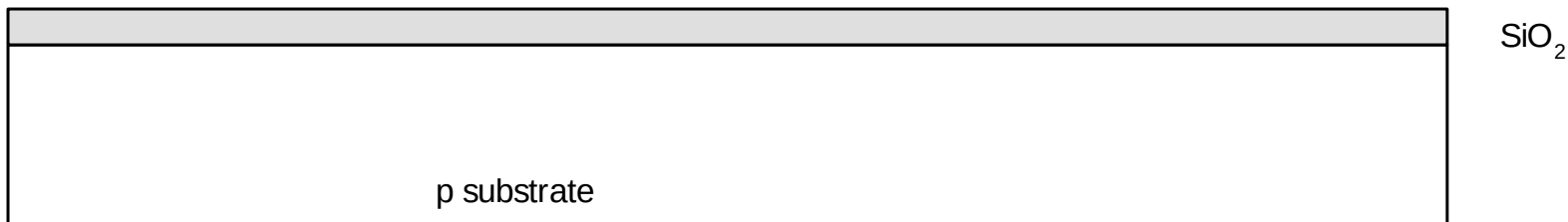


Oxidation

- Grow SiO_2 on top of Si wafer
 - 900 – 1200 C with H_2O or O_2 in oxidation furnace

2. Oxidation of wafer

The Silicon dioxide layer or oxide layer is created on the surface that protects the substrate. Silicon is one of the metals that are easily available with properties suitable for the fabrication process.

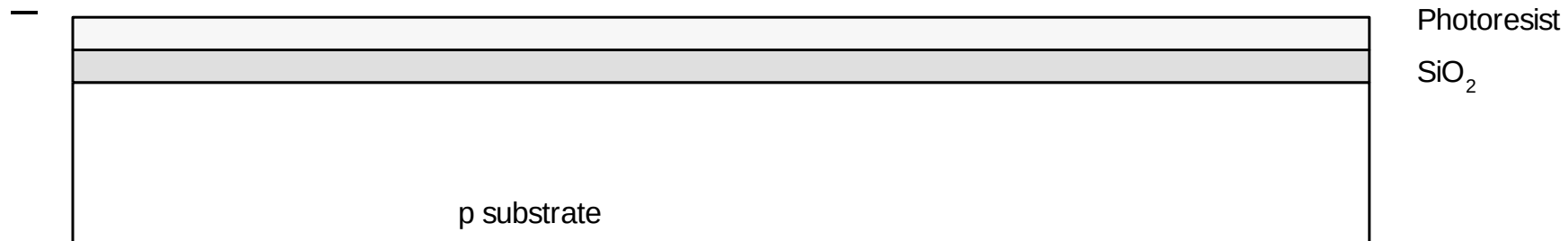


Photoresist

- Spin on photoresist
 - Photoresist is a light-sensitive organic polymer
 - Property changes where exposed to light
- Two types of photoresists (positive or negative)
 - Positive resists can be removed if exposed to UV light
 - Negative resists cannot be removed if exposed to UV light

3. Photoresist deposition

The photoresist material is deposited on the wafer. It allows the formation of the n-well on the p-type substrate.

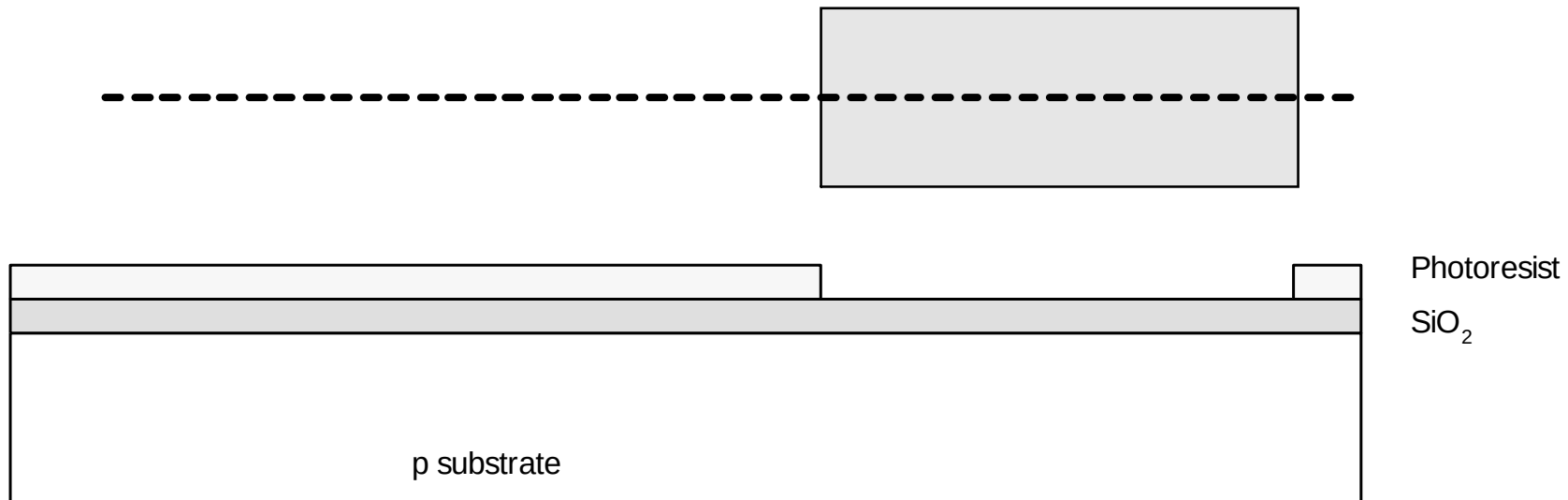


Lithography

- Expose photoresist to Ultra-violet (UV) light through the n-well mask
- Strip off exposed photoresist with chemicals

4. Lithography (for N-well Mask)

The n-well mask is exposed on the wafer with a particular pattern. The soft part or the unexposed part of the photo resist material is removed to expose the SiO_2 layer.



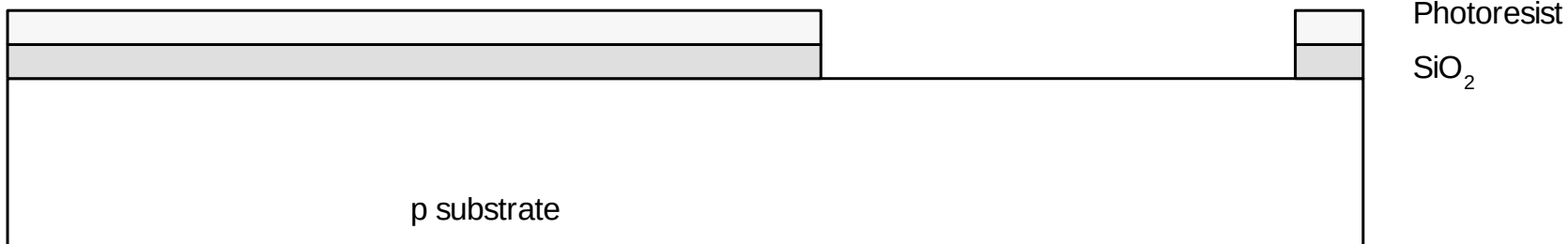
Etch

- Etch oxide with hydrofluoric acid (HF)
 - Seeps through skin and eats bone; nasty stuff!!!
- Only attacks oxide where resist has been exposed
- N-well pattern is transferred from the mask to silicon-di-oxide surface; creates an opening to the silicon surface

5. Oxide Etch

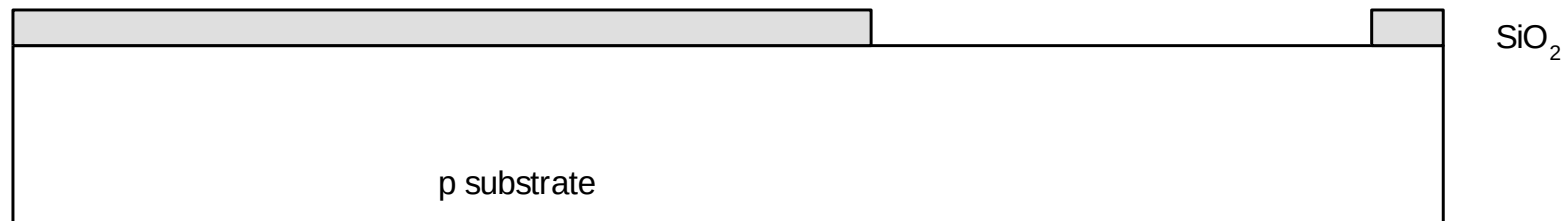
The right part of the wafer is not covered by the photoresist. It is due to the area left for the formation of the n-well. To protect it, the oxide layer is etched with the **Hydrofluoric acid**. The silicon atoms of the SiO_2 layer and the fluoride atoms of HF acid form the strong bond. The silicon wafer is now ready to be exposed to the n-well area.

The leftover photoresist material is removed with the help of the etching process.



Strip Photoresist

- Strip off remaining photoresist
 - Use mixture of acids called piranha etch
- Necessary so resist doesn't melt in next step

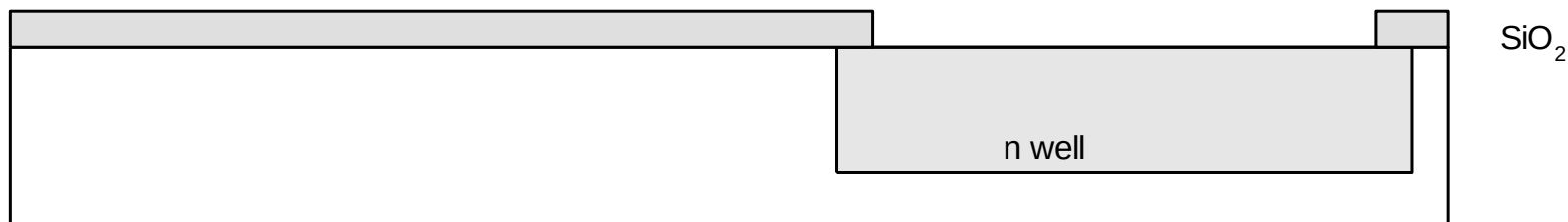


n-well

- n-well is formed with diffusion or ion implantation
- Diffusion
 - Place wafer in furnace with arsenic-rich gas
 - Heat until As atoms diffuse into exposed Si
- Ion Implantation
 - Blast wafer with beam of As ions
 - Ions blocked by SiO_2 , only enter exposed Si
- SiO_2 shields (or masks) areas which remain p-type

6. N-well formation

The diffusion process is used to make the n-well on the right side of the wafer. Diffusion is a method of adding impurities from a high concentration region towards a low concentration region. We can also use the ion implantation process to create an n-well.

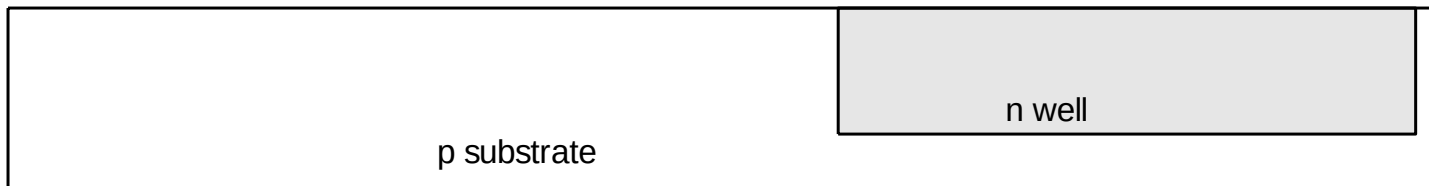


Strip Oxide

- Strip off the remaining oxide using HF
- Back to bare wafer with n-well
- Subsequent steps involve similar series of steps

7. Oxide removal

The remaining oxide on the surface of the wafer is stripped off with the help of HF acid. The oxide protects and needs to be removed to create a gate junction. It is done to expose the n-well formed in the above step directly.



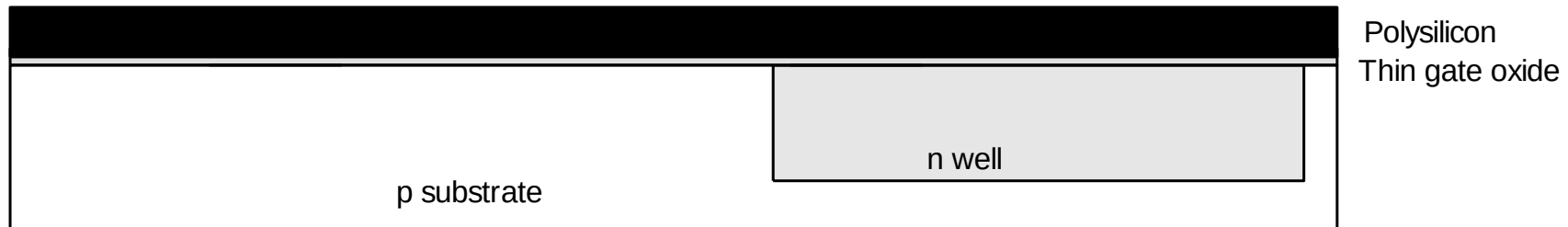
Polysilicon

(self-aligned gate technology)

- Deposit very thin layer of gate oxide
 - $< 20 \text{ \AA}$ (6-7 atomic layers)
- Chemical Vapor Deposition (CVD) of silicon layer
 - Place wafer in furnace with Silane gas (SiH_4)
 - Forms many small crystals called polysilicon
 - Heavily doped to be good conductor

8. Gate formation

The polysilicon is added to the surface that forms a gate structure deposited by CVD process. It is a heavily doped layer of polysilicon present over the thinox, which is a type of thin gate oxide.

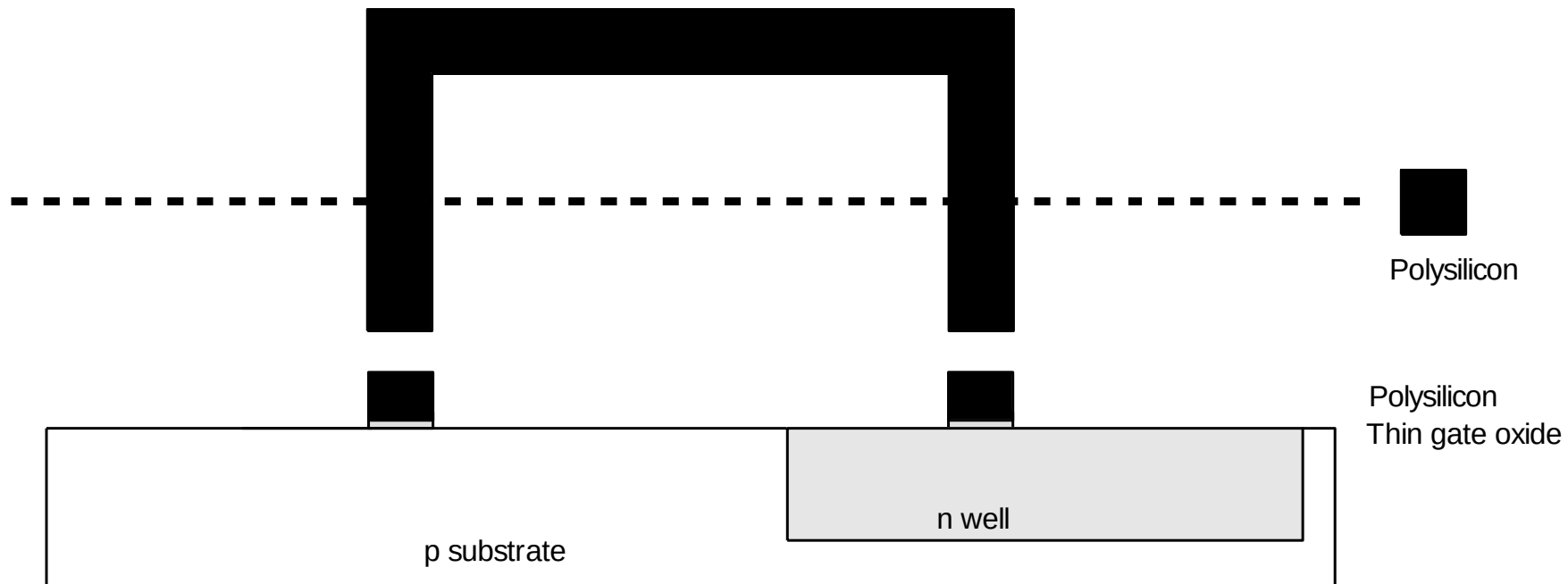


Polysilicon Patterning

- Use same lithography process discussed earlier to pattern polysilicon

9. Poly patterning

The wafer surface is exposed with the photoresist and the mask to create the two G terminals. It is because CMOS is a combination of NMOS and PMOS, and both have separate gates

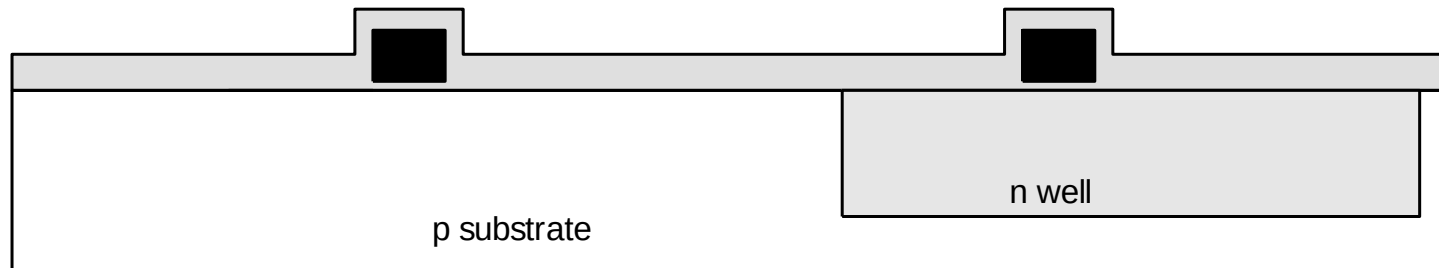


Self-Aligned Process

- Use gate-oxide/polysilicon and masking to expose where n+ dopants should be diffused or implanted
- N-diffusion forms nMOS source, drain, and n-well contact

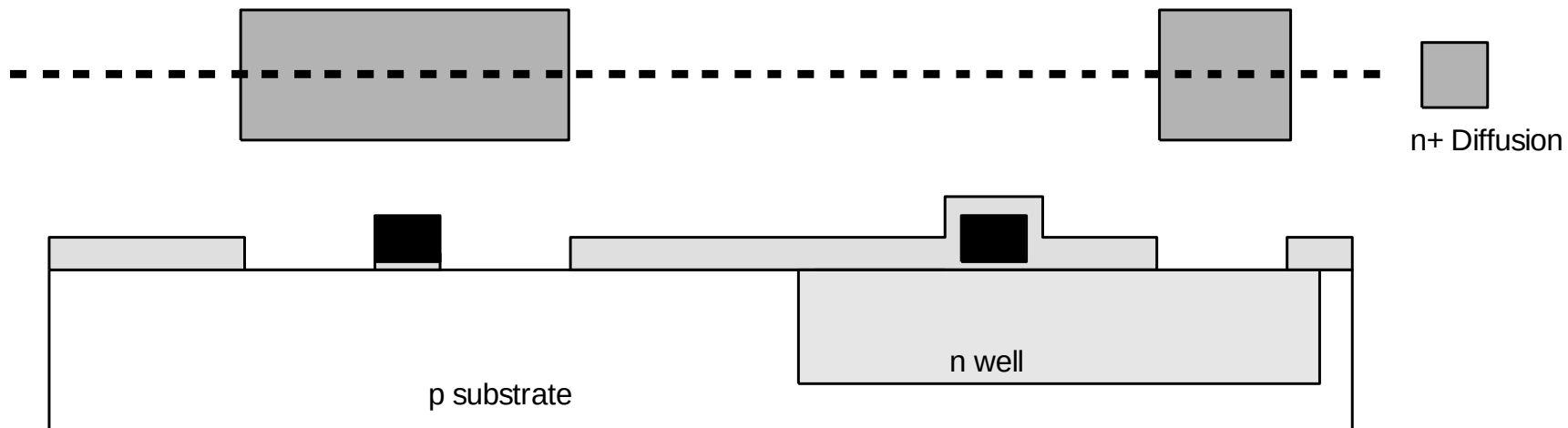
10. Diffusion pattern

A protective layer of oxide and photoresist is again applied on the wafer to protect the two gate terminals.



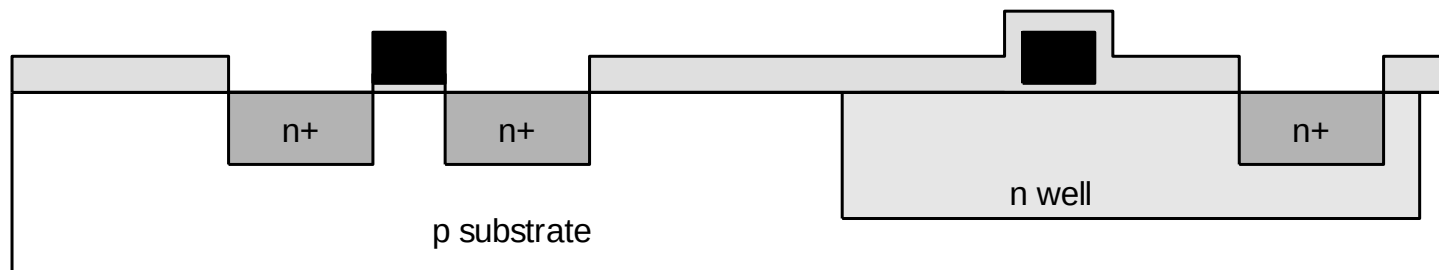
N-diffusion/implantation

- Pattern oxide and form n+ regions
- *Self-aligned process* where gate blocks n-dopants
- Polysilicon is better than metal for self-aligned gates because it doesn't melt during later processing



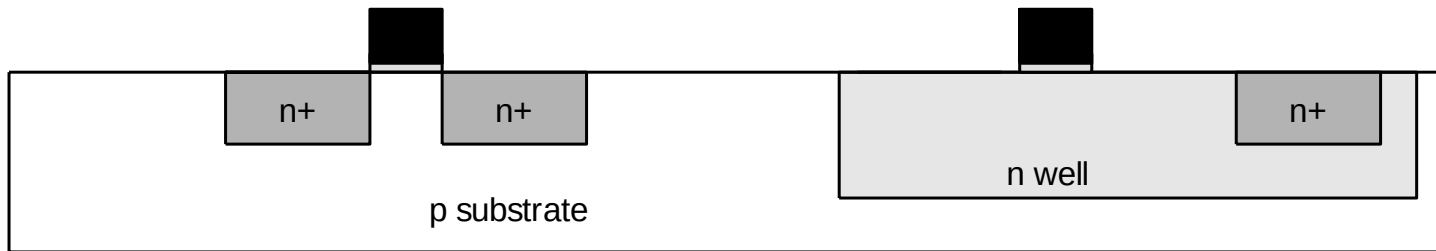
N-diffusion/implantation cont.

- Historically dopants were diffused
- Usually high energy ion-implantation used today
- But n+ regions are still called diffusion



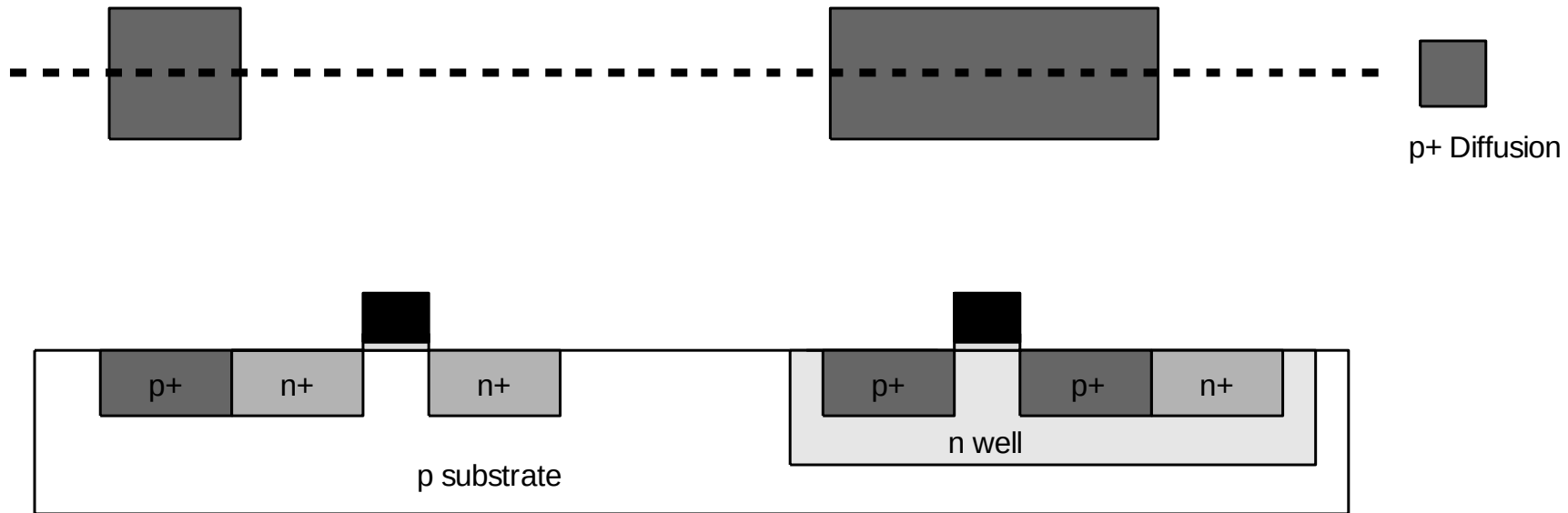
N-diffusion cont.

- Strip off oxide to complete patterning step



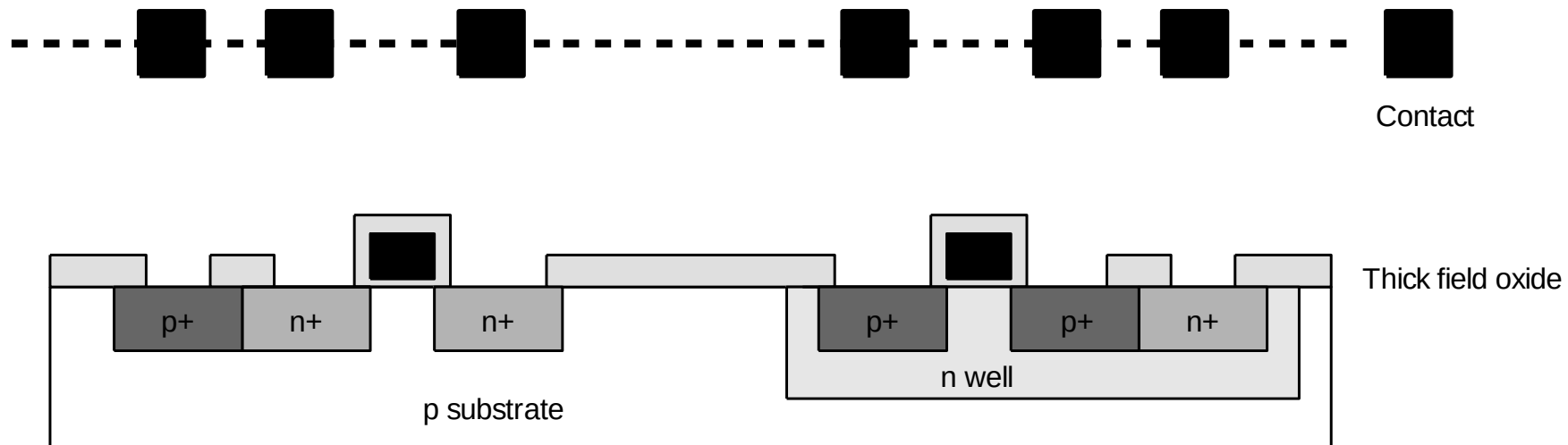
P-Diffusion/implantation

- Similar set of steps form p+ “diffusion” regions for PMOS source and drain and substrate contact



Contacts

- Now we need to wire together the devices
- Cover chip with thick field oxide (FO)
- Etch oxide where contact cuts are needed

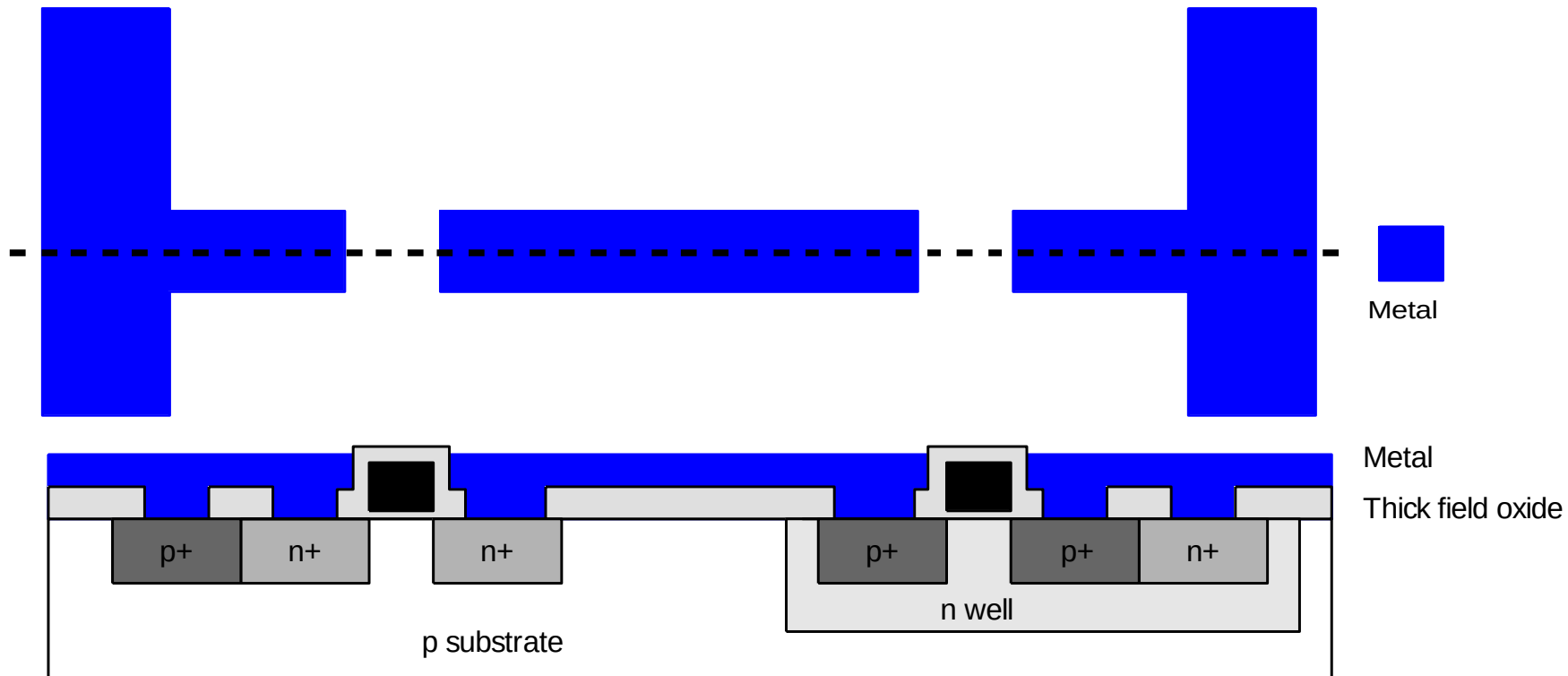


Metalization

- Sputter on aluminum over whole wafer
- Copper is used in newer technology
- Pattern to remove excess metal, leaving wires

Metal formation

The metal layer of aluminum is deposited on the surface of wafer including the contact holes. It also fills the cut holes. It is patterned with the help of metal mask.

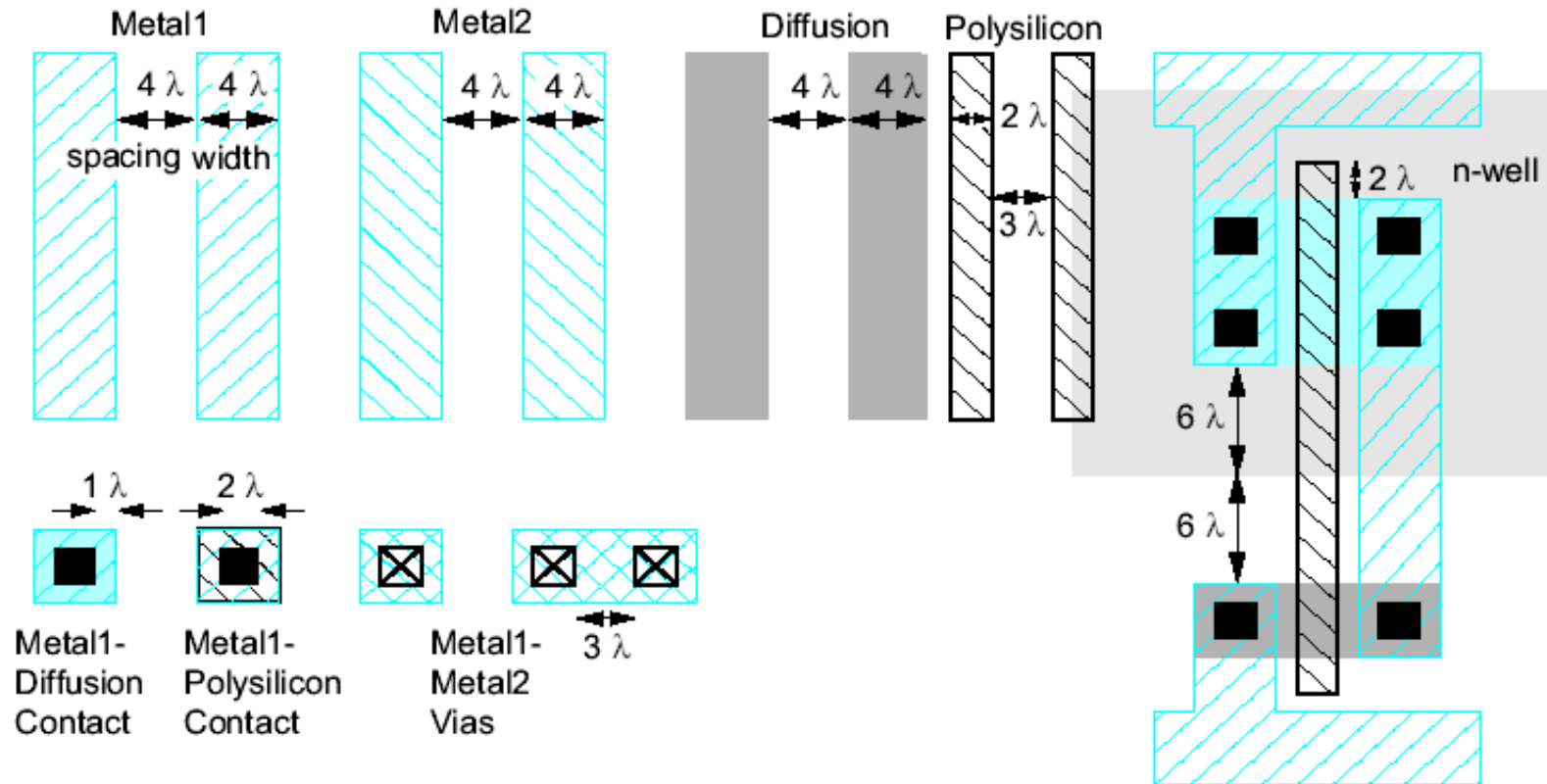


Physical Layout

- Chips are specified with set of masks
- Minimum dimensions of masks determine transistor size (and hence speed, cost, and power)
- Feature size f = distance between source and drain
 - Set by minimum width of polysilicon
- Feature size improves 30% every 3 years or so
- Normalize for feature size when describing design rules
- Express rules in terms of $\lambda = f/2$
 - E.g. $\lambda = 0.3 \mu\text{m}$ in $0.6 \mu\text{m}$ process

Simplified Design Rules

- Conservative rules to get you started



Monolithic IC

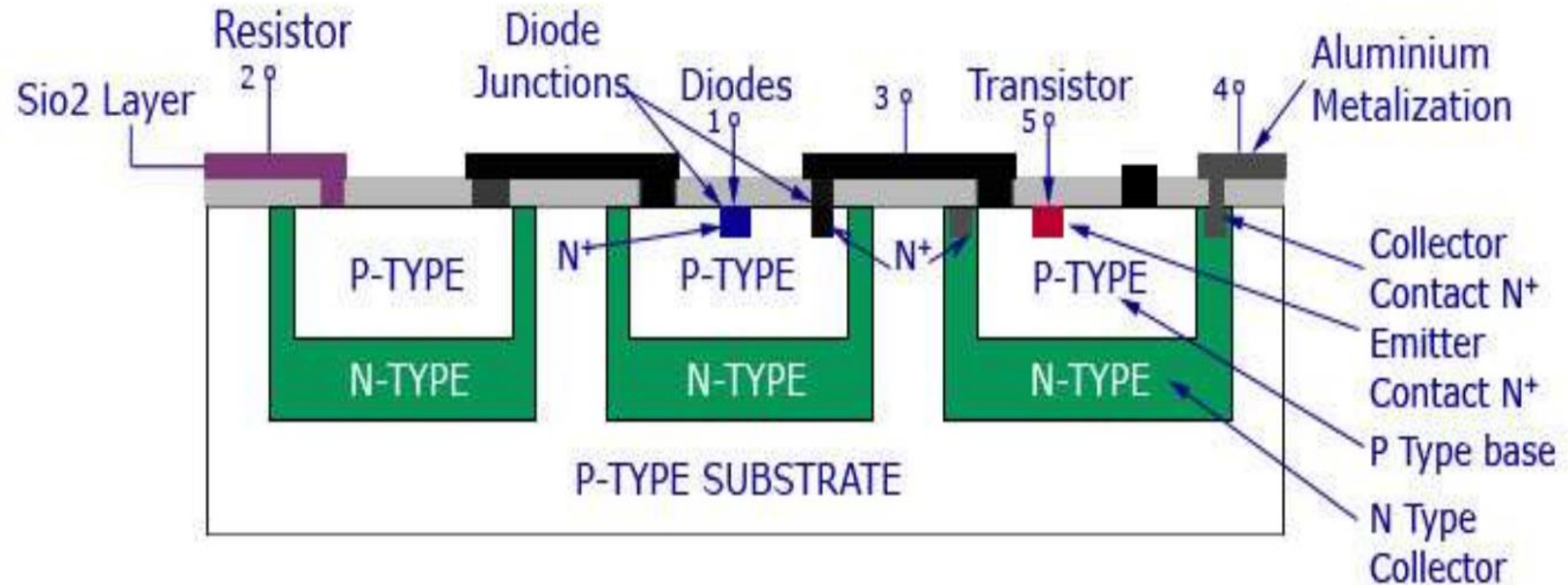
- The term monolithic is derived from Greek words monos (meaning single) and lithos (meaning stone).
- Thus a monolithic circuit is built into a single or single stone or single crystal of silicon.
- The word integrated refers to the fact that all the circuit components- transistor, diodes, resistors, capacitor and their interconnections are fabricated on the single wafer.

Manufacturing Process of Monolithic ICs

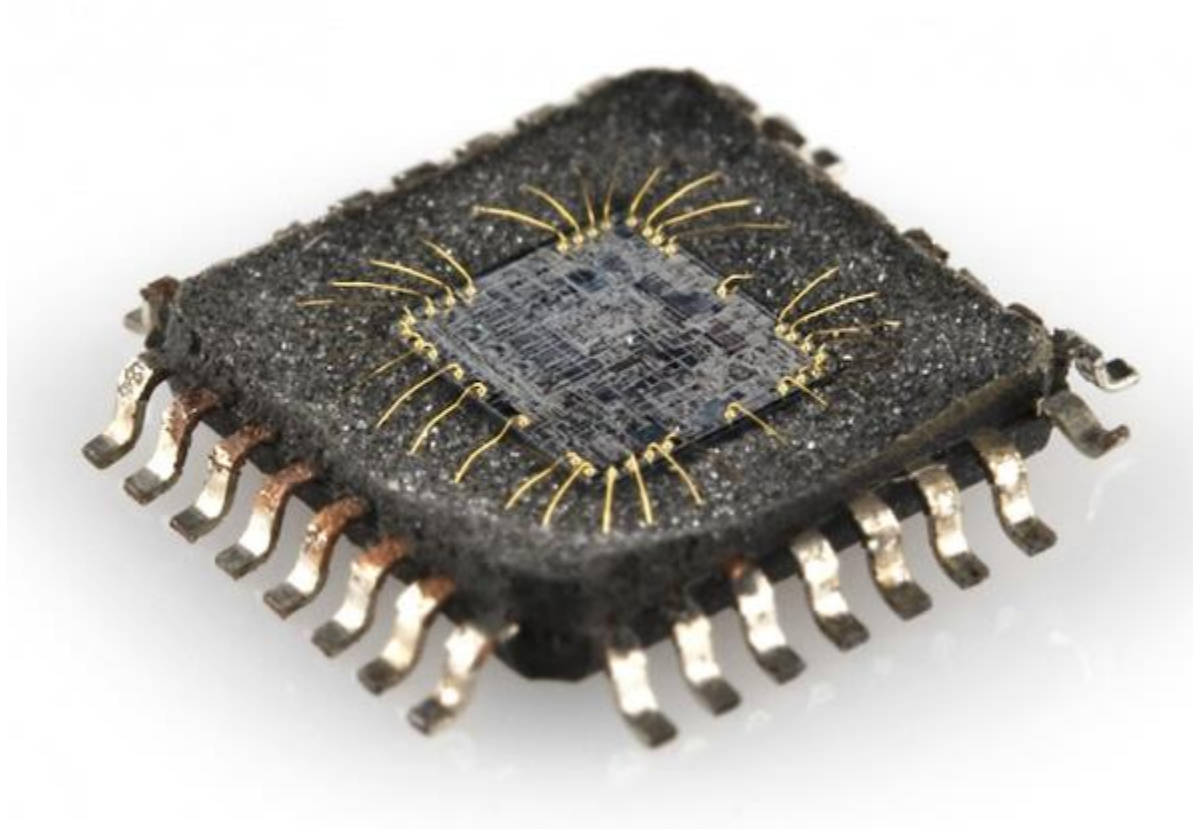
Manufacturing processes of monolithic IC using silicon planar technology can be classified as follows:

1. Silicon wafer preparation
2. Epitaxial growth
3. Oxidation
4. Photolithography
5. Diffusion
6. Ion implantation
7. Isolation techniques
8. Metallization
9. Assembling and packaging process

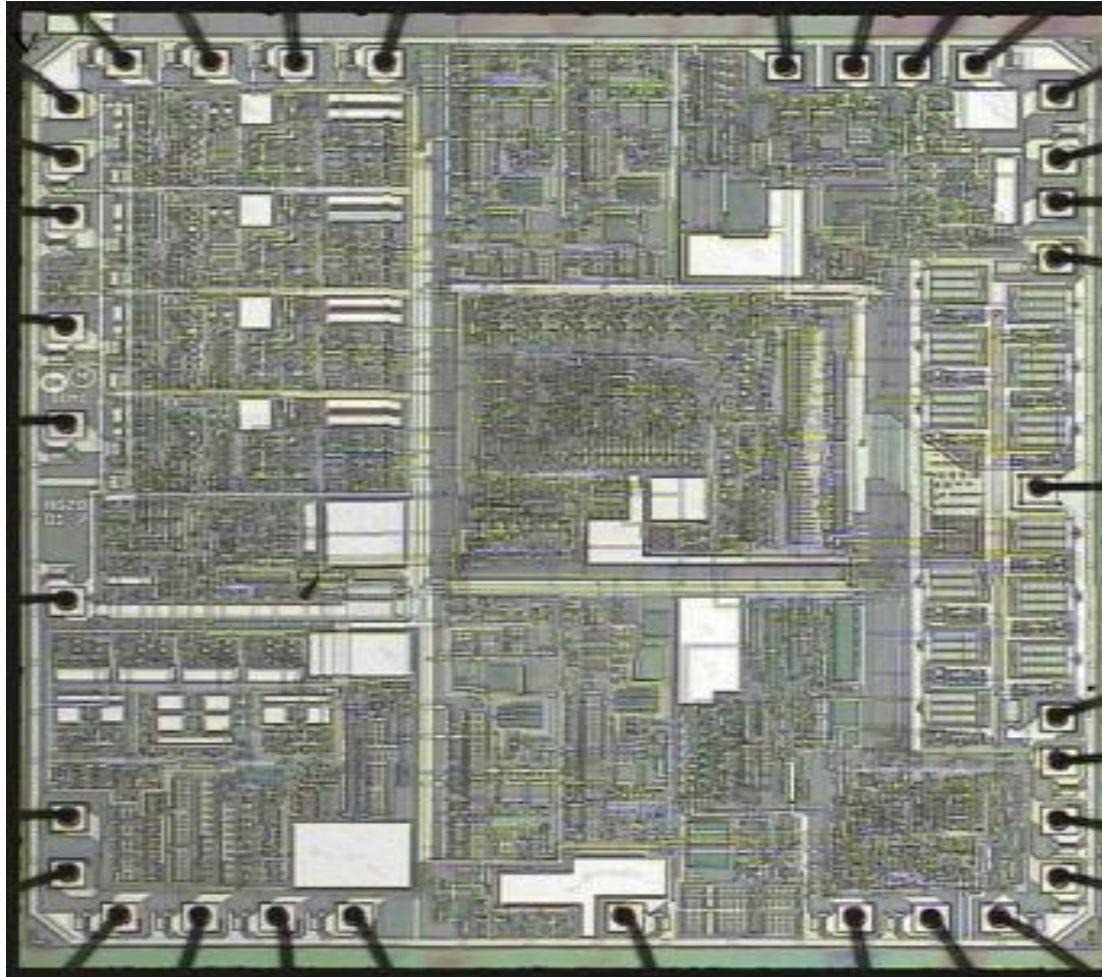
Basic Monolithic IC Cross-Sectional View



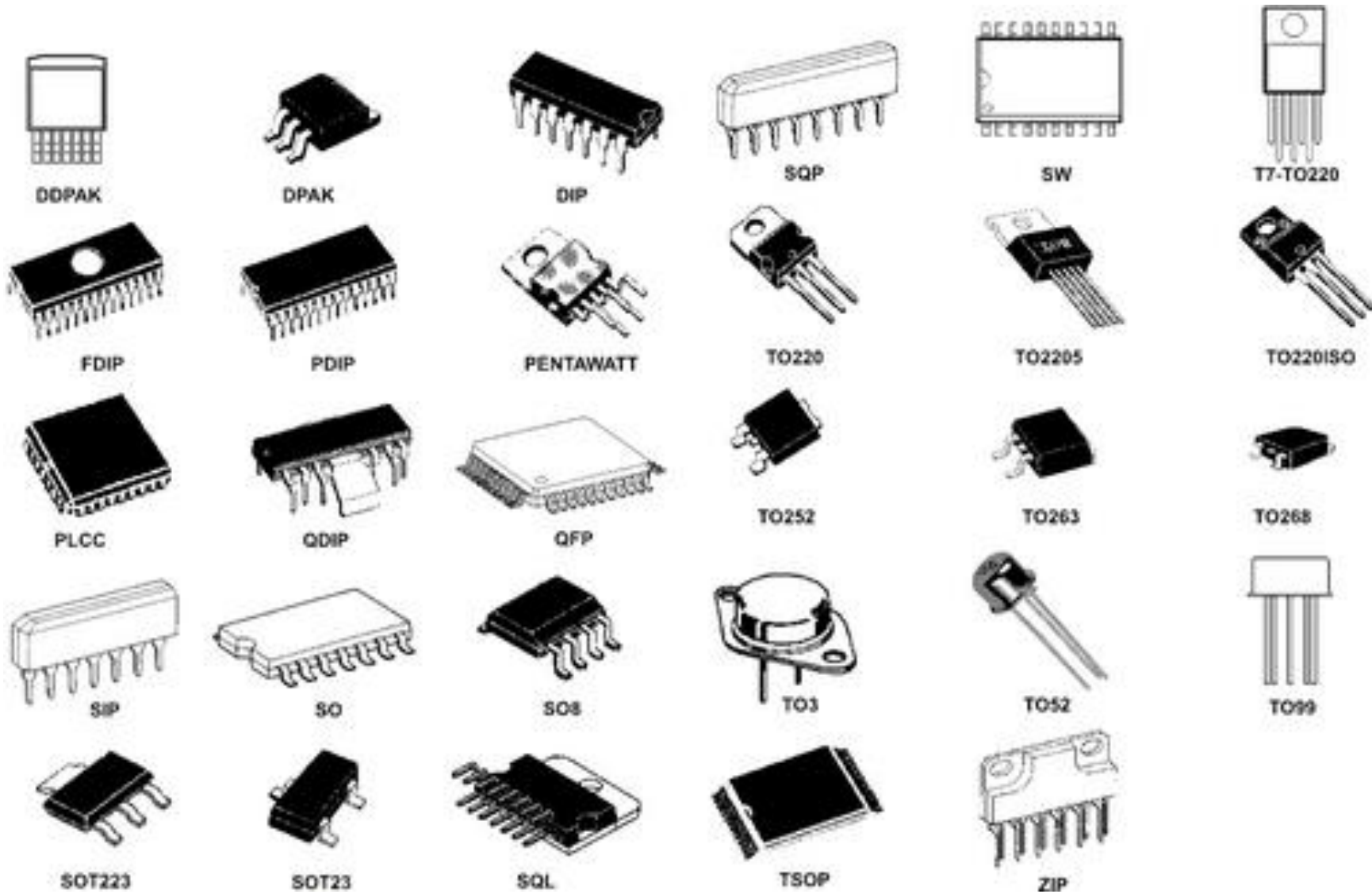
Inside the IC



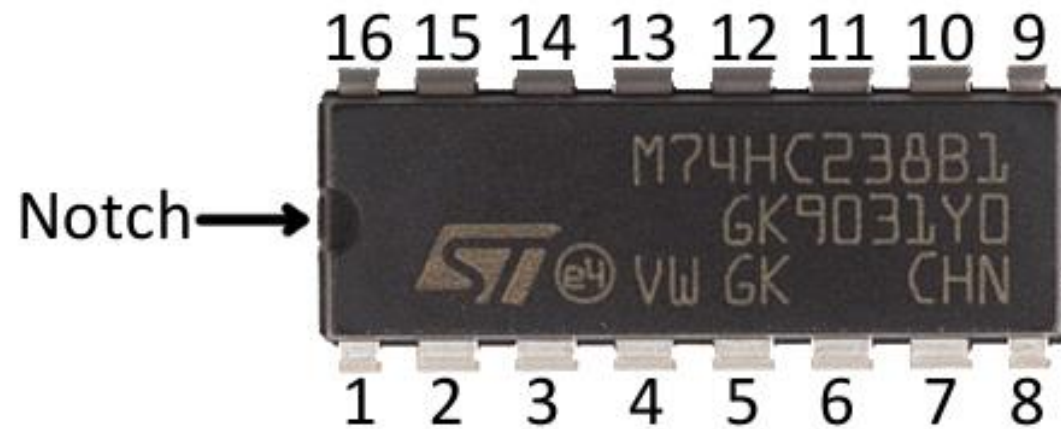
Overview of IC Die



IC Packages



Polarity Marking and Pin Numbering



Advantages of integrated circuits

1. Miniaturization and hence increased equipment density.
2. Cost reduction due to batch processing.
3. Increased system reliability due to the elimination of soldered joints.
4. Improved functional performance.
5. Matched devices.
6. Increased operating speeds.
7. Reduction in power consumption

Limitations of IC

- (i) Very small resistors (less than a few ohms) and very large resistors (of the order of meg ohm) cannot be obtained by the IC technology.
- (ii) Large capacitances (greater than 200 pF) cannot be produced.
- (iii) Inductors and transformers cannot be fabricated in the IC.
- (iv) The tolerance of the IC passive components is large (typically ± 20 per-cent).
- (v) The IC components are voltage sensitive and have large temperature coefficients.
- (vi) The performance of *p-n-p* transistor is poor.
- (vii) The power dissipation capacity is small.
- (viii) The high-frequency performance of ICs is limited by the parasitic capacitances.

With the progress in technology, some of these limitations are expected to be removed in future.